

Nibrexo OS AI

04 — Complete Feature Specification + User Roles & Permissions v1.0

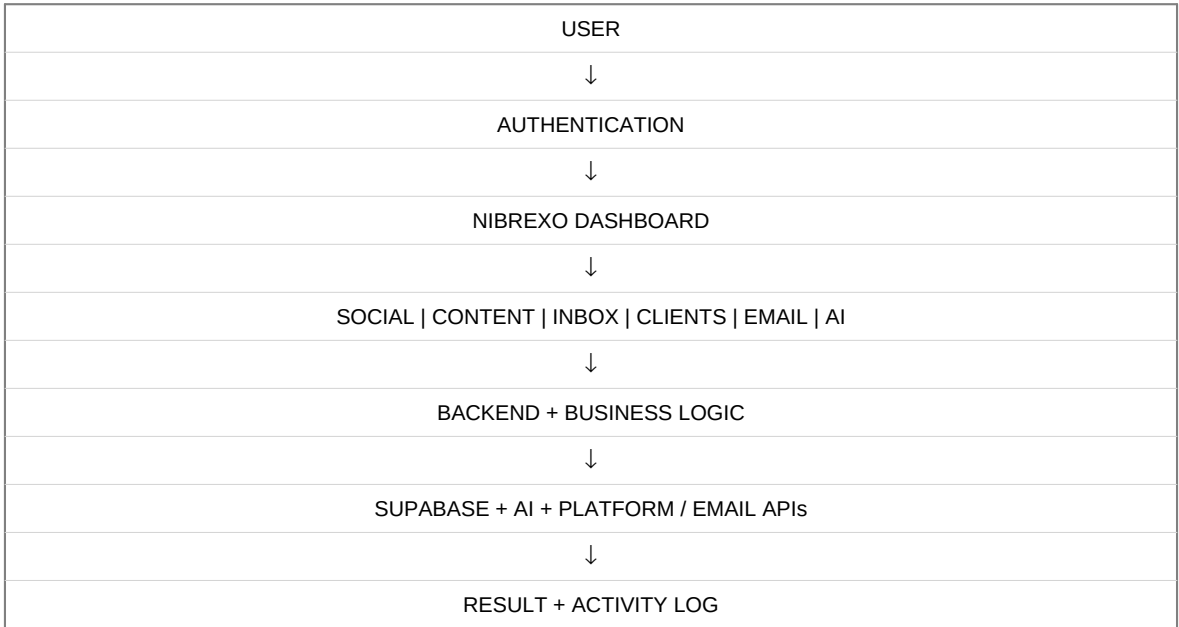
Detailed functional specification + basic flowcharts for LLM Arena / developer handoff.

Owner: Nibrexo **Status:** Working specification

1. Purpose

- Har core feature ka purpose, behavior, flow aur Definition of Done define karna.
- Coding agent ko ambiguous requirements se bachana.
- External/irreversible actions ke liye approval workflow default rakhna.
- Platform API capabilities ko implementation se pehle verify karna.

2. Product-Level Flow



3. User Roles & Permissions

Owner / Admin

- Full organization control.
- Manage users, permissions, integrations, clients and content.
- Configure automation and AI settings.
- Access critical system settings.

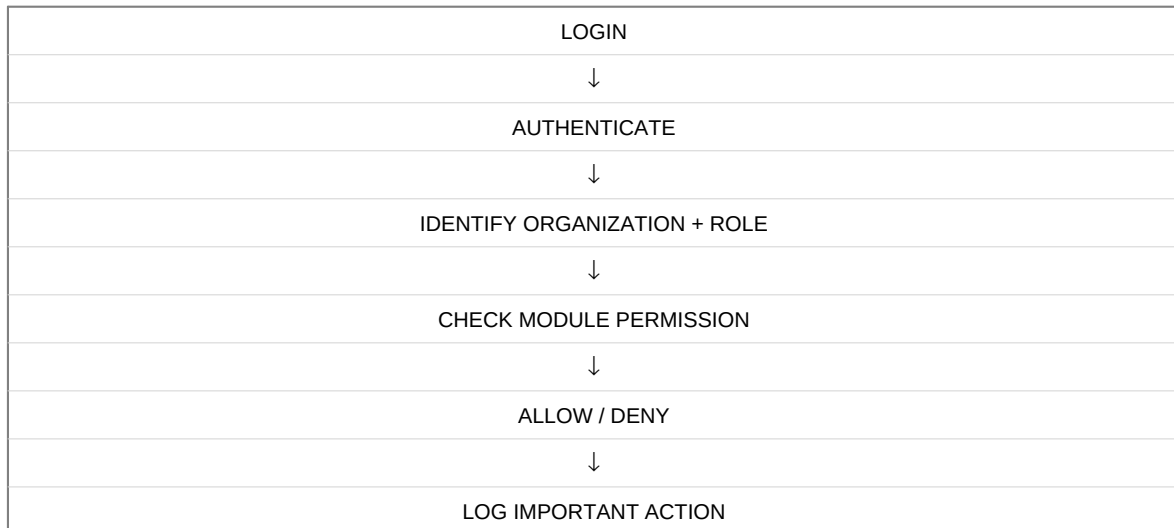
Team Member

- Access only assigned modules.
- Create/edit content and handle inbox where permitted.
- Manage assigned clients.
- Cannot change critical organization/security settings without permission.

Future Client Role

- Restricted workspace access.
- View assigned content/communication.
- Approve content/emails when enabled.
- No access to internal settings or other clients.

Permission Flow

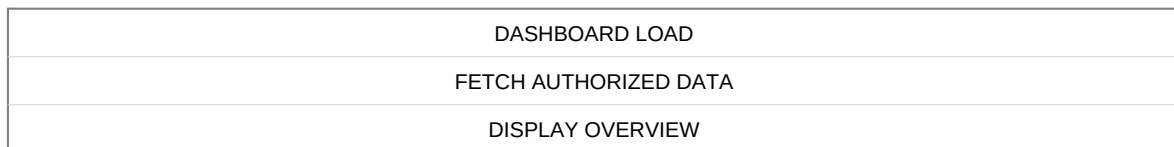


4. Dashboard

Purpose: Centralized control center.

Behavior: Overview cards, pending approvals, scheduled content, inbox alerts and email activity.

Basic Flow



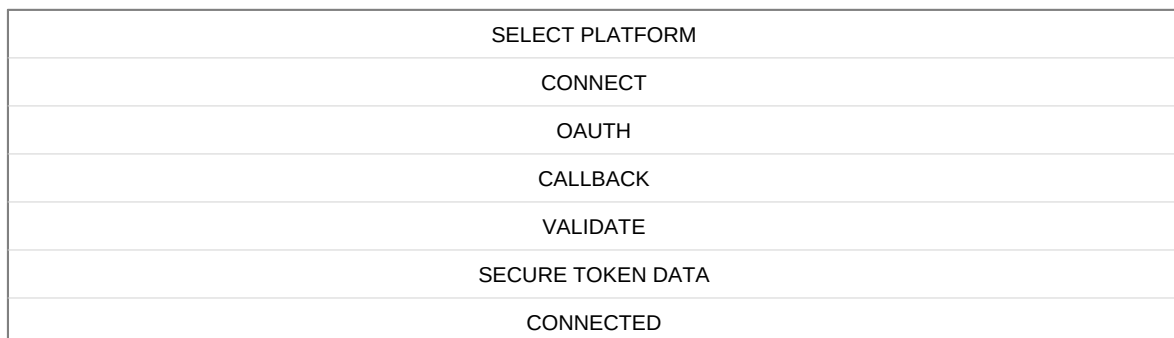
Definition of Done: Authorized data only; loading/empty/error states work.

5. Social Account Management

Purpose: Securely connect supported platforms.

Behavior: OAuth/official API authorization, connection status, reconnect and disconnect.

Basic Flow



Definition of Done: Account connects/disconnects and state is reliable.

6. Content Library

Purpose: Central media workspace.

Behavior: Upload, preview, metadata, drafts, search/filter and archive/delete by permission.

Basic Flow

UPLOAD
VALIDATE
SUPABASE STORAGE
CONTENT RECORD
LIBRARY
EDIT / ARCHIVE

Definition of Done: Media and database records remain consistent.

7. AI Content Assistant

Purpose: Speed up content creation.

Behavior: Generate captions, descriptions, hooks, CTAs and hashtags using objective/platform/tone.

Basic Flow

IDEA
PLATFORM + TONE
AI
REVIEW
SAVE / USE

Definition of Done: Output is editable and does not auto-publish without authorization.

8. Multi-Platform Publishing

Purpose: Publish/schedule selected content.

Behavior: Validate each platform separately and track independent results.

Basic Flow

CONTENT
PLATFORMS
VALIDATE
SCHEDULE / PUBLISH
PLATFORM APIS
RESULTS

Definition of Done: Per-platform status, clear failures and duplicate protection.

9. Unified Inbox

Purpose: Centralize supported communication.

Behavior: Normalize DMs/comments/mentions where supported into conversations.

Basic Flow

PLATFORM DATA
ADAPTER
NORMALIZE

UNIFIED THREAD
INBOX
FILTER / SEARCH / ASSIGN

Definition of Done: Correct platform identity and thread context.

10. AI Communication Assistant

Purpose: Generate useful reply suggestions.

Behavior: Classify intent, use context/brand voice, generate editable response.

Basic Flow

NEW MESSAGE
CLASSIFY
CONTEXT
AI REPLY
HUMAN REVIEW
SEND / EDIT / REJECT

Definition of Done: Approved reply uses supported original platform API.

11. Client CRM

Purpose: Centralize client information and activity.

Behavior: Profiles, contacts, notes, status, social links, conversations and email history.

Basic Flow

CREATE CLIENT
PROFILE
LINK ACTIVITY
TIMELINE
STATUS

Definition of Done: Client data respects permissions.

12. Email Templates

Purpose: Standardize professional emails.

Behavior: Categories, subject, body, variables, edit/duplicate/archive.

Basic Flow

CREATE
CATEGORY
SUBJECT + BODY
SAVE
USE

Definition of Done: Reusable, editable and permission-protected.

13. Client Email Sending

Purpose: Send approved client emails through provider API.

Behavior: Template + client + optional AI personalization, preview and approval.

Basic Flow

CLIENT
TEMPLATE
PERSONALIZE
PREVIEW
APPROVE
PROVIDER API
STATUS

Definition of Done: Provider-based send is logged; no local/manual sending.

14. Email Automation

Purpose: Run controlled follow-up sequences.

Behavior: Delays, stop conditions, reply/bounce handling where supported.

Basic Flow

SEQUENCE
CLIENTS
STEP 1
WAIT
CHECK
STOP / NEXT

Definition of Done: No duplicate sends; steps logged.

15. Notifications & Approvals

Purpose: Surface important events.

Behavior: Pending approvals, failed posts/emails, disconnected accounts and critical events.

Basic Flow

EVENT
CLASSIFY
NOTIFICATION
USER
APPROVE / FIX

Definition of Done: Actionable and permission-aware.

16. AI Copilot

Purpose: Coordinate multiple modules via natural language.

Behavior: Plan, permission-check, execute, verify and report multi-step tasks.

Basic Flow

COMMAND
INTENT
PLAN
PERMISSION

TOOLS
EXECUTE
VERIFY
APPROVAL
REPORT

Definition of Done: No permission bypass; failures/results explained.

17. Settings

Purpose: Manage account, organization, integrations, AI and notifications.

Behavior: Segment sensitive settings by permission and validate changes.

Basic Flow

SETTINGS
CATEGORY
AUTH CHECK
EDIT
VALIDATE
SAVE
AUDIT

Definition of Done: Sensitive changes protected.

18. Security Rules

- Never expose API keys/secrets in frontend.
- Use Supabase Row Level Security for data access.
- Server-side privileged actions require authorization checks.
- Store OAuth tokens securely; never show raw secrets in UI.
- AI cannot bypass user permissions.
- Verify supported webhook signatures.
- Validate uploaded files.
- Use rate limits/retry/backoff where needed.
- Log important external actions.

19. Edge Cases

- Disconnected account → reconnect state.
- Expired token → re-authentication.
- API outage → pending/failed with reason.
- Partial publishing → preserve each platform result.
- Duplicate request → idempotency/duplicate protection.
- Email bounce → stop/flag according to rule.
- Bad AI output → regenerate/edit, do not send.
- Insufficient permissions → deny with explanation.

- Missing media requirement → validation before publishing.

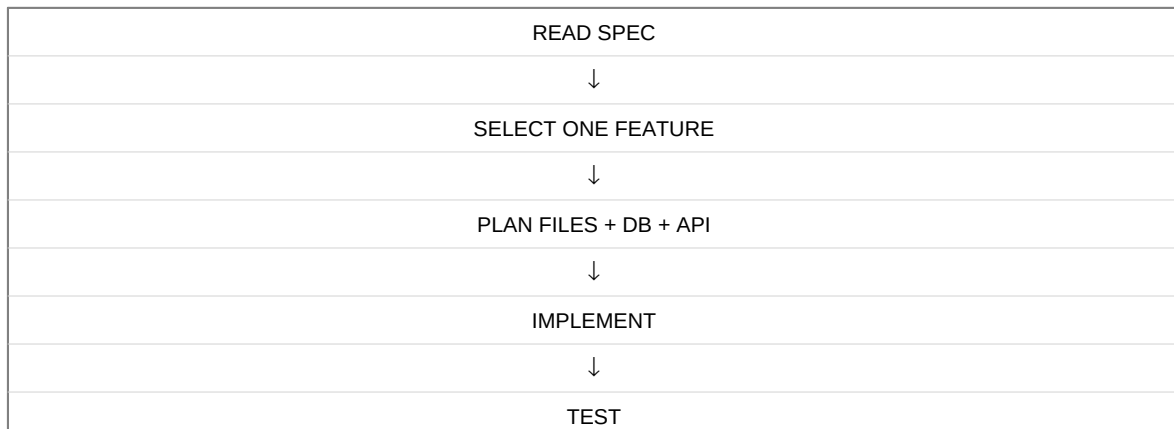
20. MVP vs Future

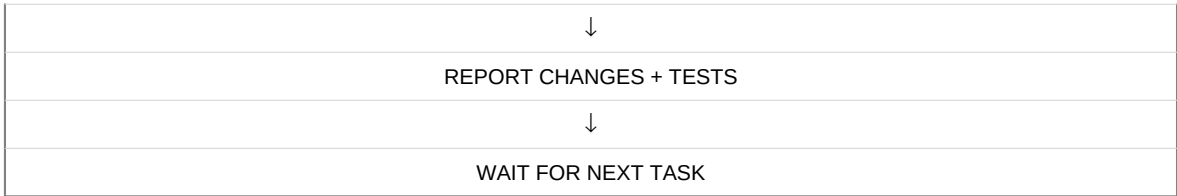
Module	MVP	Future
Auth	✓	Team/SSO
Social	✓ Selected APIs	More platforms
Publishing	✓	Advanced orchestration
Inbox	Foundation	Advanced routing
AI replies	Suggestions + approval	Controlled auto-replies
CRM	Basic	Advanced
Email	Templates + sending	Advanced campaigns
Automation	Basic sequences	Advanced workflows
Analytics	Basic	Advanced reporting
AI Copilot	Core actions	More autonomous workflows

21. Standard Definition of Done

- Requirements documented.
- Database reviewed.
- Backend/API complete.
- Frontend complete.
- Permissions enforced.
- Errors/edge cases handled.
- Tests pass.
- No secrets committed.
- Logging added where required.
- User acceptance complete.
- Documentation updated.
- GitHub commit/tag created.

22. LLM Arena Coding Flow





Agent must not silently expand scope, replace the approved stack, add unnecessary third-party automation services, or invent unsupported platform capabilities.